Data Center Visit

Visit 1: [Fast NUCES CFD Campus]
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Server Rooms REPORT

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Data Center A: FAST University
A brief overview

In the dynamic landscape of data management, the significance of data centers has surged exponentially. The data center at FAST National University's CFD Campus exemplifies modern data management practices. Equipped with advanced servers, networking infrastructure, and state-of-the-art cooling mechanisms, it ensures seamless operations. Meticulously organized server rooms with redundant power supplies and backup systems underscore its commitment to reliability. Advanced monitoring tools and stringent security protocols safeguard sensitive data, aligning seamlessly with international standards and best practices.

- The report delves deeply into the realm of International Standards for data centers.
- It aims to elucidate the evolving benchmarks shaping the industry's landscape.

- Focus on exhaustive evaluation of FAST National University's CFD Campus Data Center.
- Symbolizes the convergence of cutting-edge technology and robust infrastructure.

Data Center Infrastructure and Technology

DATA CENTER

"A data center is a centralized facility housing computer systems, servers, storage, networking equipment, and other components to manage, process, and store vast amounts of data, serving as the backbone of the networking industry."

Infrastructure & Facility Standards:

- **EN 50600:** This series of standards covers various aspects of data center design, construction, and operation. It includes:
    - Building construction (EN 50600-2-1)
    - Power distribution (EN 50600-2-2)
    - Environmental control (EN 50600-2-3)
    - Telecommunications cabling infrastructure (EN 50600-2-4)
    - Security systems (EN 50600-2-5)
    - Management and operational information systems (EN 50600-2-6)
- **Uptime Institute Tier Certifications:** These rank data centers based on their level of redundancy and fault tolerance, with tiers ranging from:
    - Tier I: Basic capacity with minimal redundancy.
    - Tier II: Redundant capacity components for critical systems.
    - Tier III: Concurrently maintainable with no downtime during maintenance.
    - Tier IV: Fault-tolerant with the highest level of redundancy.

Operational Standards:

- **ISO 9001:** Focuses on Quality Management Systems, ensuring consistent processes and continuous improvement.
- **ISO 14001:** Guides Environmental Management Systems, promoting resource efficiency and sustainability.
- **ISO 27001:** Sets the standard for Information Security Management Systems, protecting sensitive data.
- **PCI DSS:** Payment Card Industry Data Security Standard, crucial for organizations handling credit card information.

Additional Important Standards:

- **Uptime Institute Operational Sustainability:** Focuses on data center operational practices beyond Tier certification.
- **AMS-IX Data Centre Business Continuity Standard:** Outlines best practices for disaster recovery and business continuity.

DATA CENTER LOCATION

FAST National University of Computer and Emerging Sciences CFD Campus

BANDWIDTH

The data center boasts a robust 1.25 Gigabits per second (Gbps) bandwidth capacity. This impressive total is comprised of significant contributions from various providers: 1 Gbps from PTCL, 200 Mbps from COMSATS, and 50 Mbps from PERN. This diverse range of bandwidth sources ensures consistently high-speed and reliable network connectivity.

SERVERS

The data center houses a total of 14 servers, comprising three distinct Dell PowerEdge models:

- **Dell PowerEdge R720:** Equipped with dual Intel Xeon processors, boasting a massive 768 GB memory capacity and internal storage expandable up to a staggering 32 TB.
- **Dell PowerEdge R730:** Also featuring dual Intel Xeon processors, this model offers a memory capacity of up to 3 TB and several internal storage options, including 16 x 2.5" drives for up to 29 TB or 8 x 3.5" drives for up to 64 TB.
- **Dell PowerEdge R740:** Powered by two 2nd Generation Intel Xeon Scalable processors, this server boasts a memory capacity of up to 3 TB and flexible storage options, including up to 122.88 TB via 16 x 2.5" drives or 128 TB via 8 x 3.5" drives.
- **Dell PowerEdge R750:** The powerhouse of the bunch, featuring up to two 3rd Generation Intel Xeon Scalable processors with 40 cores each. This model offers impressive memory support, ranging from 2 TB RDIMM to 8 TB LRDIMM with speeds up to 3200 MT/s. Additionally, it boasts up to 16 Intel Persistent Memory 200 series slots, pushing the maximum memory capacity to a remarkable 8 TB.

FIREWALLS

- **Unwavering Security:** The data center boasts a robust security infrastructure with 6 firewalls.
- **Hardware Powerhouse:** Two of these firewalls are advanced Fortinet 601e hardware models.
- **Software Defense:** The remaining four firewalls operate on the Cisco ASA platform, providing a dynamic layer of protection.
- **Top-Tier Performance:** Both hardware and software firewalls are renowned for their high performance and network security capabilities.
- **Web Security Champions:** These firewalls actively defend against web-based threats, ensuring comprehensive protection.

Bandwidth Division at Firewall

- **Granular Bandwidth Control:** Policy-based routing on firewalls enables strategic bandwidth allocation.
- **Research Prioritization:** The 50 MB PERN bandwidth is exclusively dedicated to research activities, ensuring smooth access to resources like IEEE and HSC websites.
- **Optimized Entertainment Access:** Popular platforms like YouTube and Facebook are channeled through the high-capacity 1000 MB PTCL bandwidth, maximizing streaming and browsing experience.

NETWORK INFRASTRUCTURE

Wireless routers

The data center is made of strong network infrastructure, with more than 125 routers from the renowned company RUCKUS. These routers play a vital role in establishing and maintaining a high performance network connectivity and networking environment.

Switches

The data center boasts a robust network foundation built on a diverse range of Cisco switches:

- **Cisco 3750:** Layer 3 switch known for scalability, ease of management, and quality of service.
- **Cisco 3560:** Layer 3 switch designed for medium to large networks, offering enhanced security features.
- **Cisco 6509:** Modular Layer 3 switch providing flexibility and scalability through various modules.
- **Cisco 3064:** Layer 3 data center switch specifically optimized for low latency and high throughput applications within data centers.

Wireless controllers

The data center contains two wireless controllers, specifically the RUCKUS SmartZone (SZ) 100 and SZ-144. The RUCKUS SZ-100 is recognized for its scalability, efficiently supporting a significant number of access points within the data center. Whereas the SZ-144 builds upon this scalability, providing an enhanced capacity suitable for extensive and densely populated environments. This ensures optimal wireless connectivity and management.

Cabling

The cabling mechanism of the university consists of both fiber and copper cables. To interconnect the buildings of the university, an underground fiber network has been laid. Whereas, within the buildings the cables are installed in trunks where trunks are available, while the rest are installed in the ceilings.

BACKUP MECHANISM

Data Backup Schedule:

- Manual backups occur daily at midnight and weekly.

Uninterrupted Power Supply (UPS):

- Online UPS ensures seamless power supply to critical servers during outages.
- 5KV UPS provides initial backup for 5 minutes.
- 30KV UPS takes over with NARADA AGM and Lithium batteries, lasting for 30 minutes.

Diesel Generator:

- Starts up within 15 minutes, well within the UPS backup window.

VULNERABILITES

The data center lacks fire protection infrastructure, leaving the data center vulnerable to significant damage in case of potential fire hazards.

COST

- **Server Costs:**
  - 14 servers at 50,000 PKR each, totaling approximately 700,000 PKR.
  - Hard drive costs vary depending on specific capacities chosen (8TB, 4TB, 1TB) and priced at 50,000 PKR each.
- **Firewall Infrastructure:**
  - Initial cost of 50 lakh PKR.
  - Renewal cost estimated between 35-40 lakh PKR.

- **Total Estimated Cost:**
    - Ranges from 1.185 crore PKR to 1.24 crore PKR, encompassing servers, hard drives, and firewall costs.
    - Actual costs may fluctuate due to specific details and market changes.

CONCLUSION

From the collected information about the data center, it is evident that the data center is aligning with many of the recognized international standards. The data center's special emphasis on security infrastructure, firewall and policy routing for bandwidth implies a commitment to information security. These practices align with the principles of ISO/IEC 27001, International Standard for Information Security Management.

Data Center B: GC University FSD
A brief overview

The Government College University Faisalabad (GCUF) boasts a well-equipped data center designed to provide robust IT infrastructure and ensure secure data storage and processing. Meticulously organized server rooms with redundant power supplies and backup systems underscore its commitment to reliability. Advanced monitoring tools and stringent security protocols safeguard sensitive data, aligning seamlessly with international standards and best practices.

- This report comprehensively explores modern data management practices in prominent data centers.
- It sheds light on the significance and evolution of data centers in the dynamic landscape of data management.

- It meticulously examines the infrastructure, operations, and security measures of selected data centers.
- These data centers symbolize the fusion of advanced technology and robust infrastructure, playing crucial roles in their domains.

DATA CENTER LOCATION

The Government College University Faisalabad is a public university located in Faisalabad, Punjab, Pakistan.

BANDWIDTH

The GCUF data center boasts a remarkable bandwidth capacity of 1264 Megabits per second (Mbps). This translates to a significant amount of data transfer capability, ensuring smooth operation and efficient processing within the data center. This robust bandwidth is further enhanced by its diverse sources, including contributions from PTCL (1 Mbps), COMSATS (200 Mbps), and PERN (350 Mbps). This combination guarantees reliable and high-speed network connectivity, effectively supporting the data center's various functions.

SERVERS

The data center has three categories of 8 servers installed:

1. **High-Performance Server:**
   - This category likely consists of a single server with a high memory capacity of 512GB. This server is suitable for demanding tasks requiring significant processing power and memory resources.
2. **Mid-Range Servers:**
   - This category consists of 2 servers with a memory capacity of 128GB each. These servers offer a good balance of performance and cost, making them suitable for various tasks with moderate resource requirements.
3. **Physical Servers:**
   - This category likely refers to 2 physical server machines, possibly with varying memory capacities or additional features compared to the previous categories.

FIREWALLS

The main campus network of GCUF is further secured by the implementation of 2 firewalls. These firewalls act as a crucial layer of defense, monitoring and controlling incoming and outgoing network traffic. By analyzing data packets, they identify and block potential threats, safeguarding the campus network from unauthorized access, malware, and cyberattacks. This additional security measure contributes to a more secure and reliable network environment for students, faculty, and staff.

Bandwidth Division at Firewall

The exact division of bandwidth between the firewalls wouldn't be a straightforward split based on their number. It's likely a strategic allocation designed to optimize network performance, security, and redundancy within the GCUF main campus network.

NETWORK INFRASTRUCTURE

Wireless routers

The GCUF network boasts a robust infrastructure with approximately 140 routers. These routers belong to two prominent brands: Ruckus and Huawei. This diverse mix of brands suggests a strategic approach to network management, potentially leveraging the strengths of each brand to optimize performance, functionality, and cost-effectiveness across the network.

Switches

The data center has

Approx. 20 buildings X 2(per building) = 40

Approx. 18 Labs X 4(per building) = 72

Total=112

It's a combination of Cisco switches (3750, 3560, 6509, 3064) employed throughout the network, potentially supplemented by other models from Ruckus and Huawei depending on specific requirements and functionalities needed in different areas.

Wireless controllers

The GCUF's data center boasts two RUCKUS SmartZone controllers: the SZ-100 and SZ-144. The SZ-100 excels in scalability, effectively managing numerous access points, while the SZ-144 offers even greater capacity for extensive campus-wide coverage, ensuring optimal wireless connectivity and management.

Cabling

Installation Method:

Digging: Trenching is the primary method for laying cables underground. This ensures protection and durability for the network infrastructure.

Cable Types:

- **Cat-6 UTP (Unshielded Twisted Pair):** This is the most likely cable type used extensively due to its cost-effectiveness and suitability for most network applications.
- **Fiber Optics:** This high-bandwidth cable is likely used for critical connections or long distances where high data transfer rates are required.
- **Limited STP (Shielded Twisted Pair) and FTP (Foiled Twisted Pair):** These shielded cables are used in specific areas where electrical interference is a concern.

BACKUP MECHANISM

Data Backup Schedule:

- Manual backups occur daily .

Diesel Generator:

- Starts up within 15 minutes, well within the UPS backup window.

VULNERABILITES

The data center lacks fire protection infrastructure, leaving the data center vulnerable to significant damage in case of potential fire hazards.

COST

Total Investment: 30 Crore PKR (as of 2022)

By incorporating the high-end estimate for UPS & Generator, the total cost of the GCUF data center falls within the 30 Crore PKR range

- Servers: 14 servers at 50,000 PKR each = 7,00,000 PKR
- Routers: 140 routers at 22 lakh each = 30,80,00,000 PKR
- Switches: 112 switches at 18 lakh each = 19,92,00,000 PKR
- Firewall & Controller: 50 lakh initial cost + 35-40 lakh renewal = 85-90 lakh PKR (approximate)
- UPS & Generator:75-85 lakh

Servers + Routers + Switches + Firewall/Controller = 51,79,00,000 PKR

Functionality of servers

DNS, FILE, Active, proxy, web server

Total Number of users and directory

Total approx. 20,000 users and an active directory

Intelligent access points

Intelligent access points are implemented because of their efficiency and benefits.

Tagging

Yes, Tagging is implemented otherwise it would be impossible to remember so many departments.

Steps for emergency pov

- Inside backup and outside backup. Outside the data center on the cloud.
- If wire/shark cut down then the solution is implemented by satellites solution

Isp service provider

1) PTCL
2) HEC

Modem

Modulation and demodulation. Yes, in the form of light electric signal transformation is happening

Gateways & Load balancers

- Gateways focus on security and access control.
- Load balancers prioritize efficient traffic distribution and high availability.

GCUF's data center utilizes both gateways and load balancers for comprehensive network management and optimization.

Standards of data center

Standards Include:

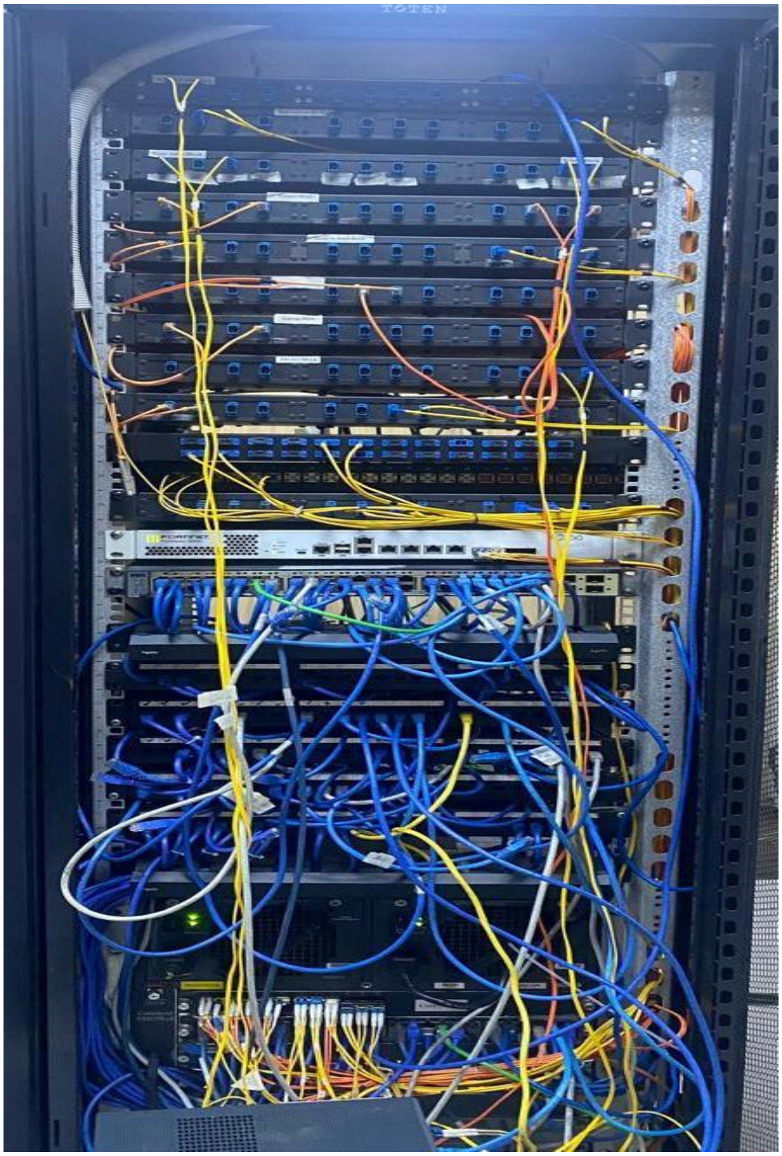
- ISO/IEC 27001: Information Security Management
- ANSI/TIA-942: Telecommunications Infrastructure Standard for Data Centers
- ASHRAE TC 9.9: Thermal Guidelines for Data Processing Environments
- Uptime Institute Tier Standard: Data Center Tier Classification System

GCUF’S Data center fulfils all the standard but **Not Certified due to Cost Constraints**

CONCLUSION

In conclusion, the data center at Government College University Faisalabad (GCUF) stands as a testament to the institution's commitment to modern data management practices. While not certified due to financial constraints, the center effectively implements industry standards to ensure reliability, security, and efficiency in its operations. With a focus on maximizing resources and optimizing performance, GCUF's data center plays a pivotal role in supporting the academic and administrative functions of the university, embodying the convergence of technology and infrastructure in the realm of data management.

GCUF Site Photos

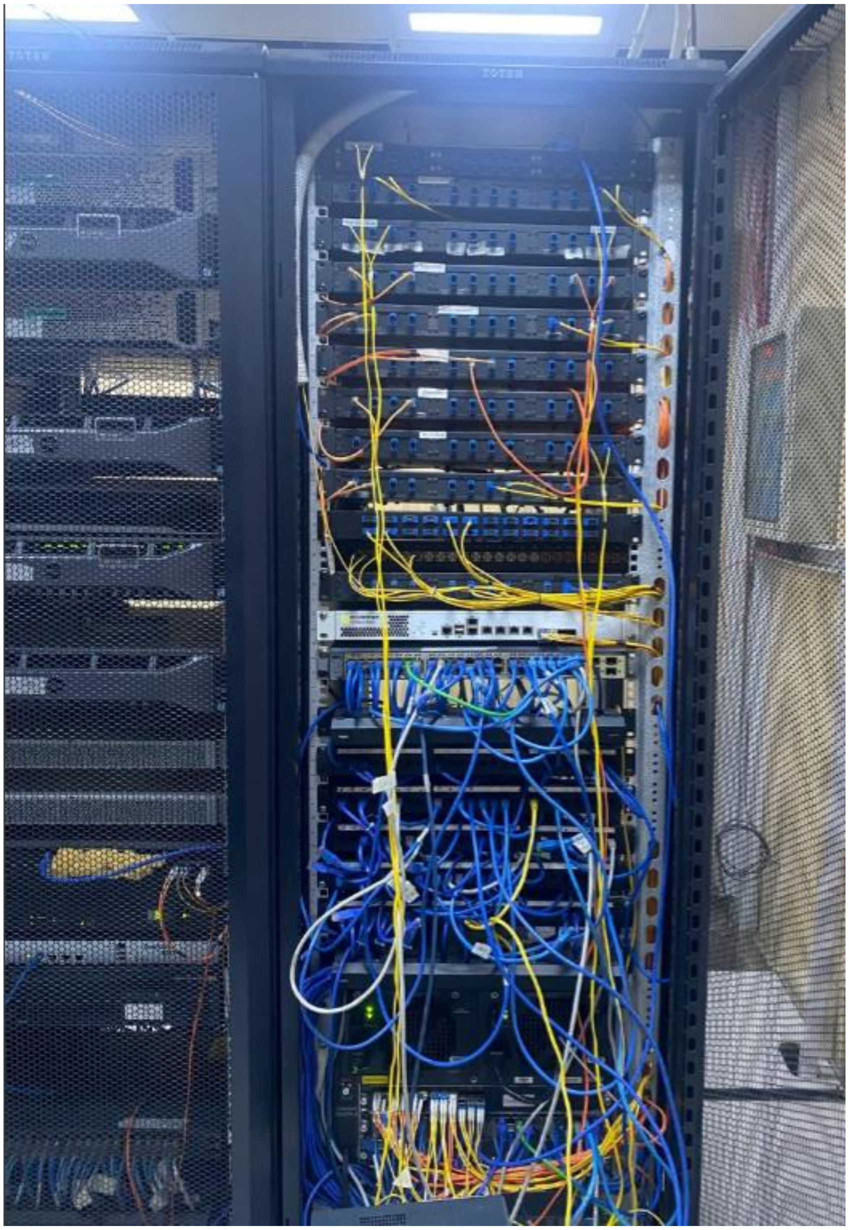


Server Racks

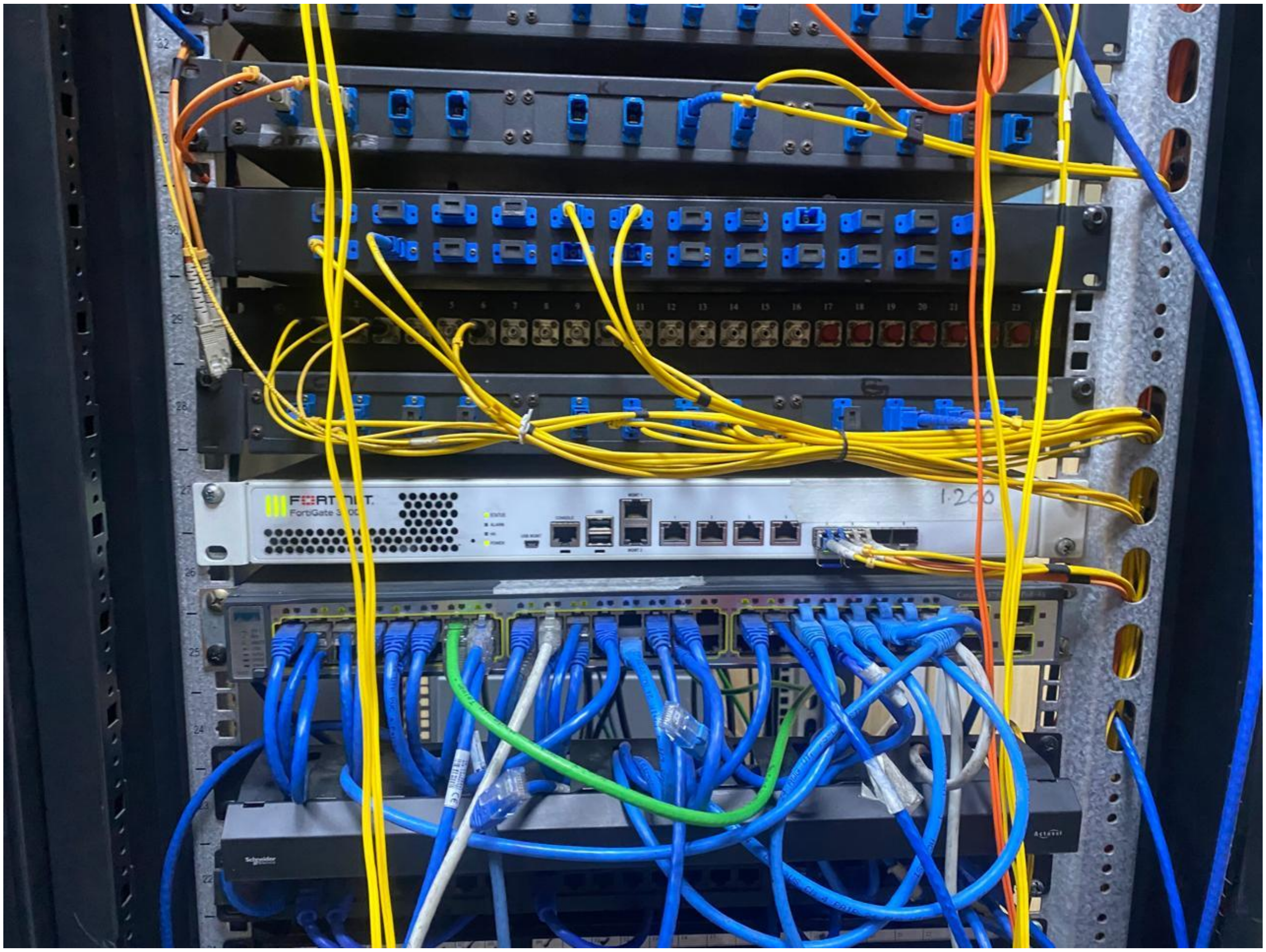

Basic Science Block
Takbeer Block
Quaid -e- Azam Block
Zoology Block
Physics Block

Tagging

Cabinet, physical setup of the computer system with multiple monitors and devices connected to it.



Nicely organized wires



Power Cables & Data Cables

Common data cable types used in server racks include:

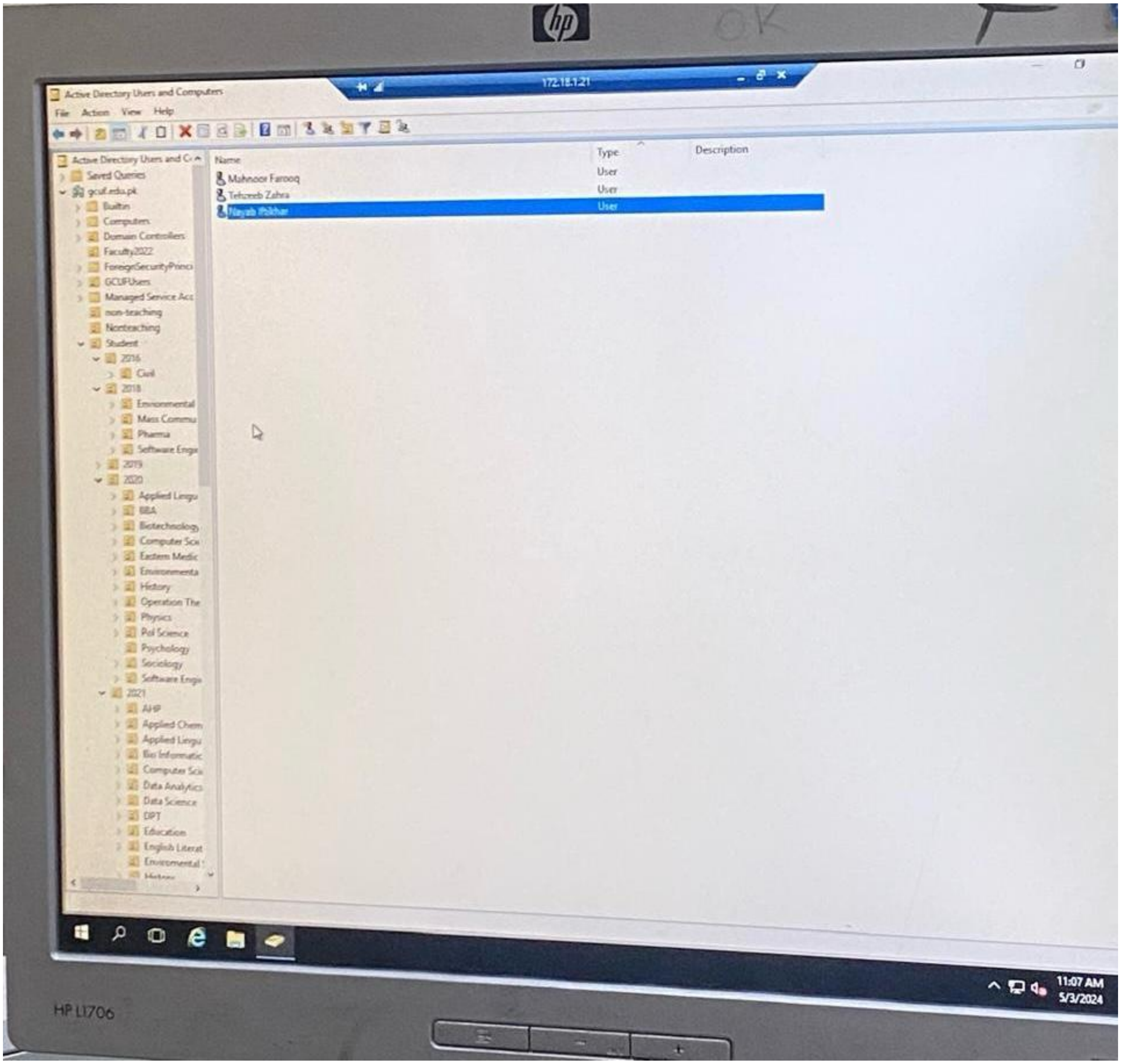
- Cat5e or Cat6 Ethernet cables for connecting servers to a network switch.
- Fiber optic cables for transmitting data over long distances or at high speeds.


S5 131
S4 137
DELL
S3 132
S2 135
DELL
HIKVISION
S1
DELL
S0
DELL
PERN Router

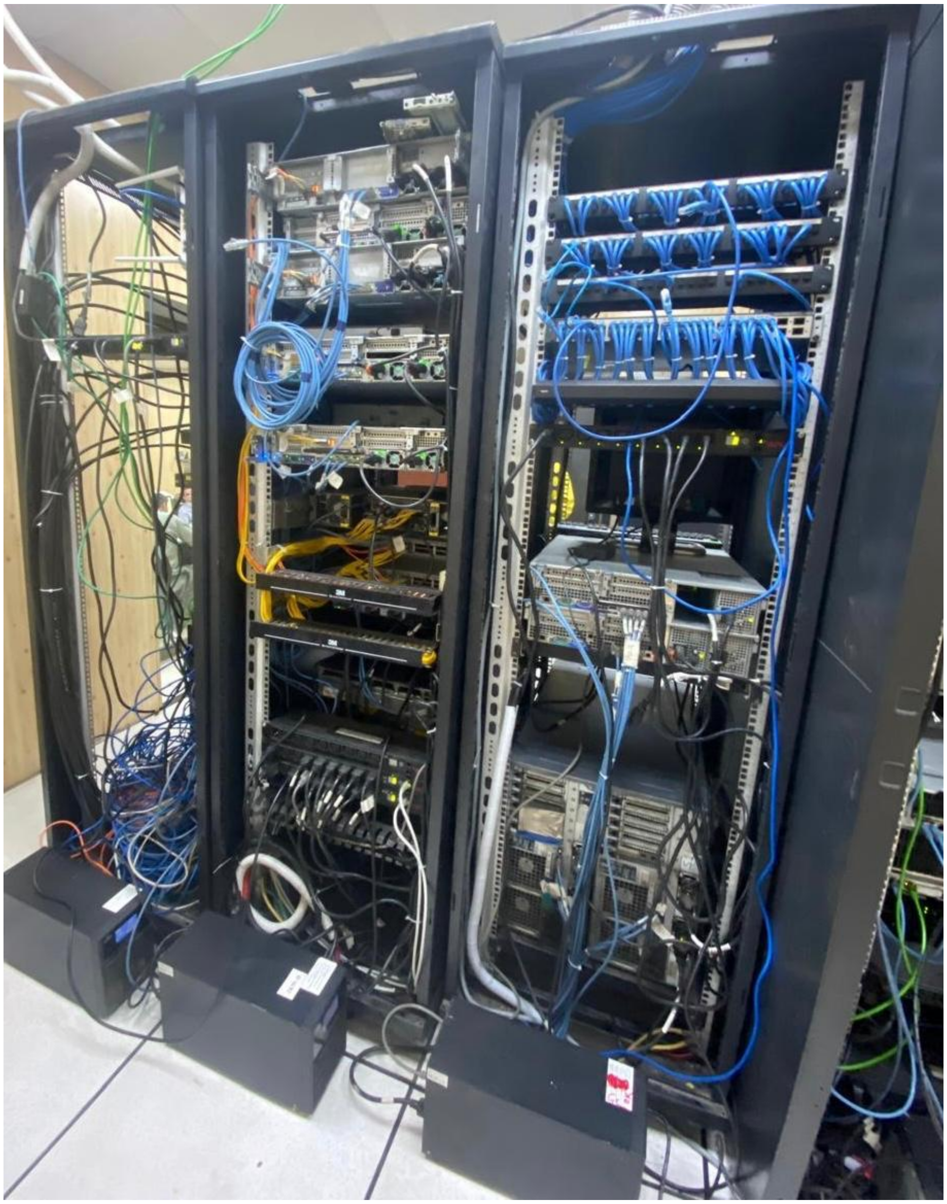
Servers

Fortinet 300D **Next-generation firewall (NGFW)**

Cisco System **Cisco routers and switches**



Active Directory Users and Computers



Server Connectivity and Cooling

These ports include:

- **Network ports**
- **Power supply connections**
- **Console ports**

Data Center